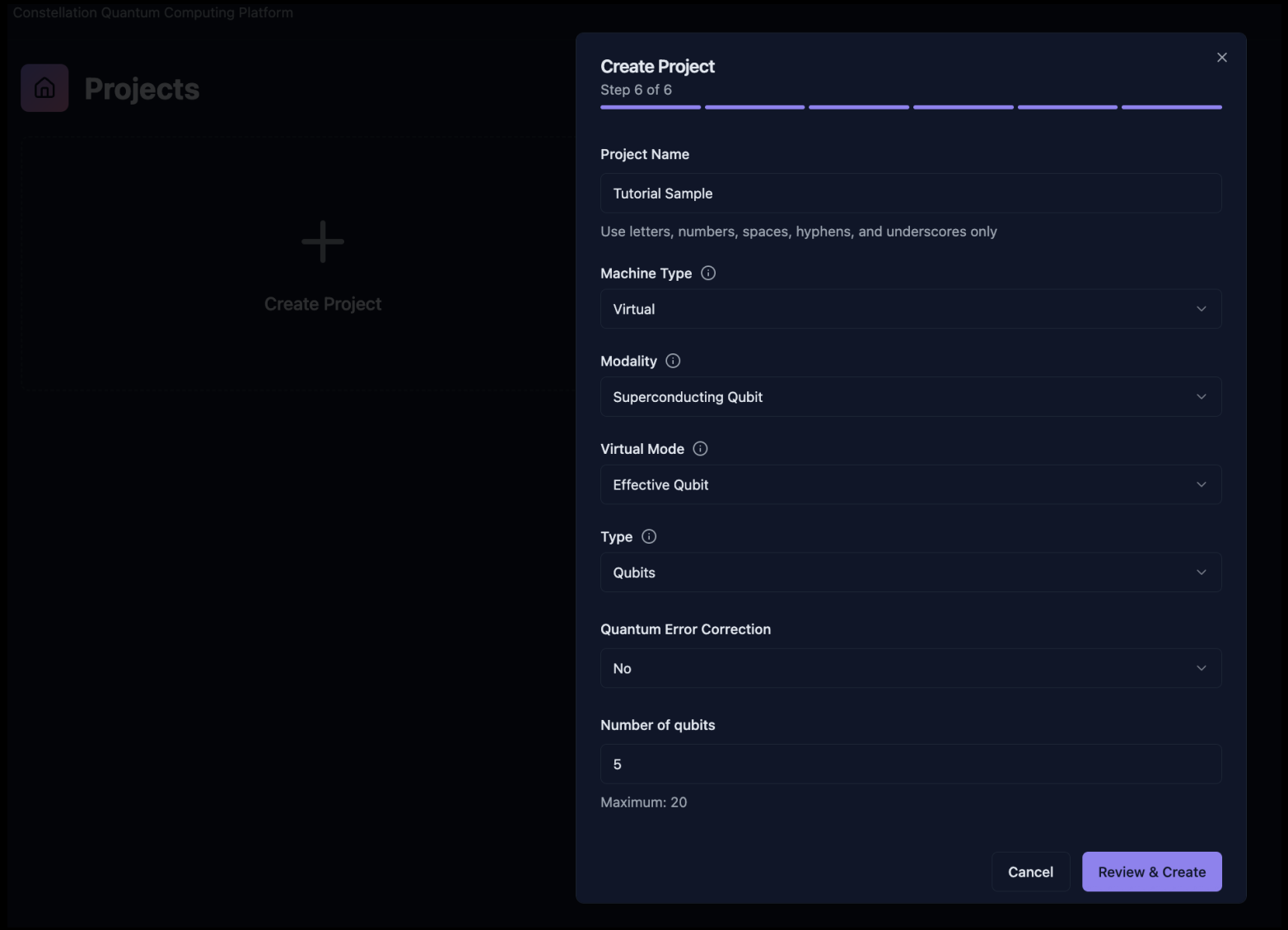
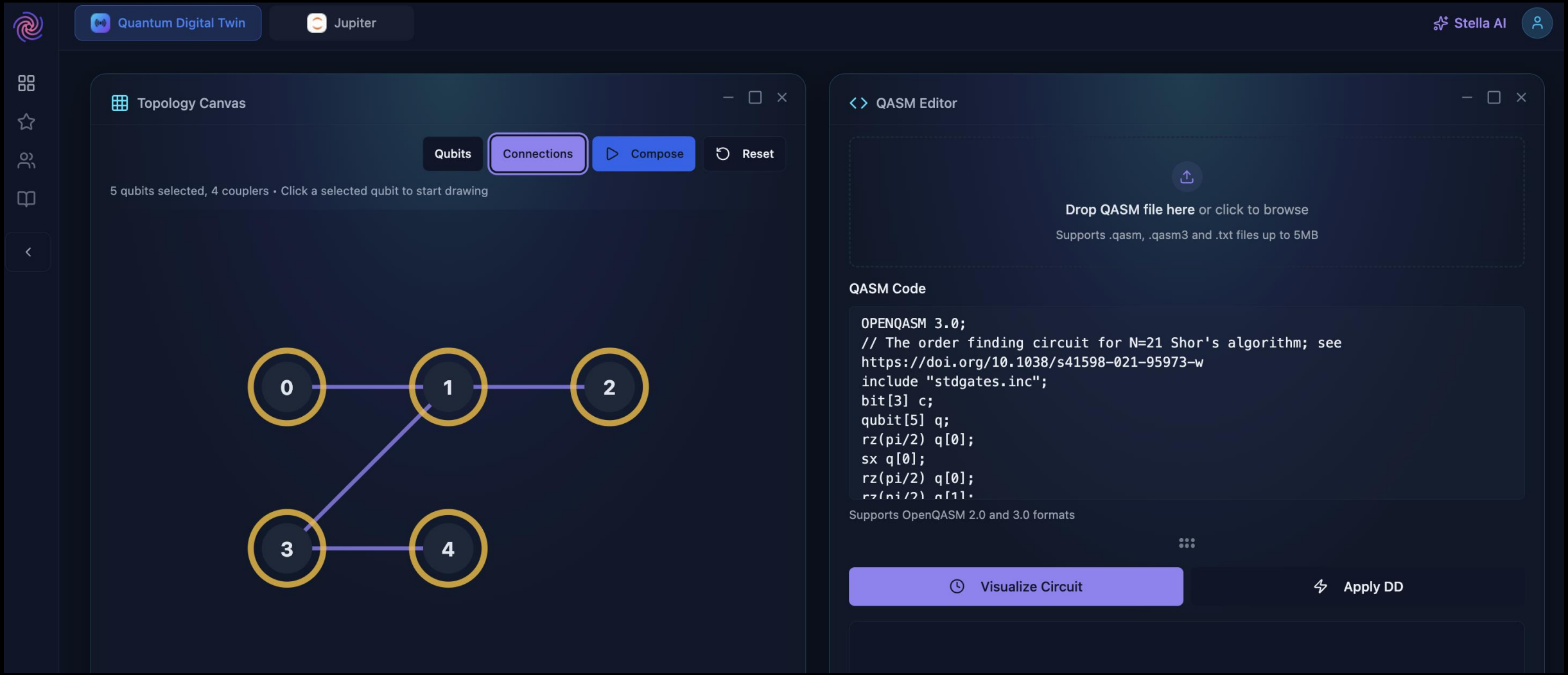
Tutorial Sample: Improve Shor’s algorithm with dynamical decoupling

Create project with a 5-qubit superconducting digital twin



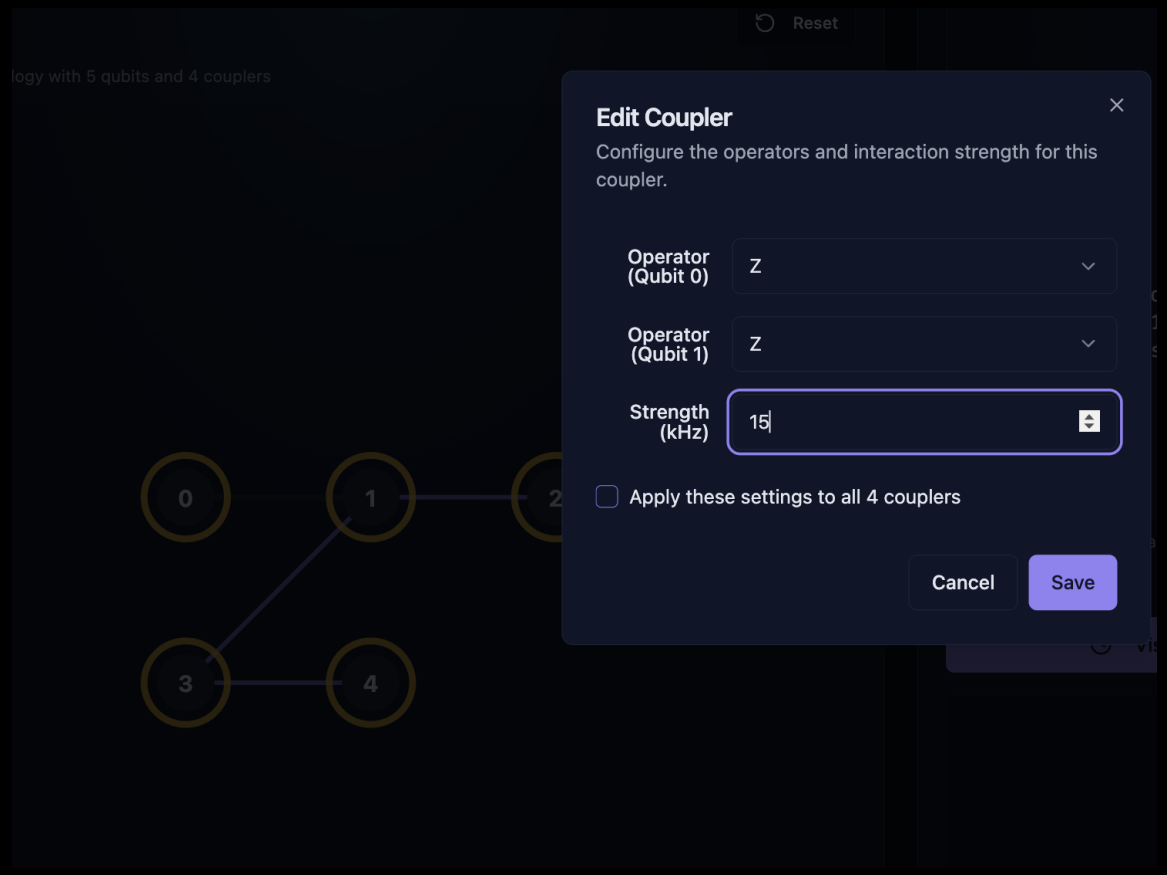
Tutorial Sample: Improve Shor’s algorithm with dynamical decoupling

Design qubit layout and connectivity in Topology Canvas



Tutorial Sample: Improve Shor’s algorithm with dynamical decoupling

Edit qubit and coupler properties



Tutorial Sample: Improve Shor’s algorithm with dynamical decoupling

Upload the QASM circuit for the order-finding algorithm and apply dynamical decoupling

QASM Code

```
OPENQASM 3.0;
include "stdgates.inc";
bit[3] c;
qubit[5] q;
rz(pi/2) q[0];
sx q[0];
rz(pi/2) q[0];
rz(pi/2) q[1];
sx q[1];
rz(pi/2) q[1];
```

Supports OpenQASM 2.0 and 3.0 formats

Visualize Circuit | Apply DD

Circuit Visualization 5 qubits 343 gates

100% Hide idle qubits Run

QASM Code

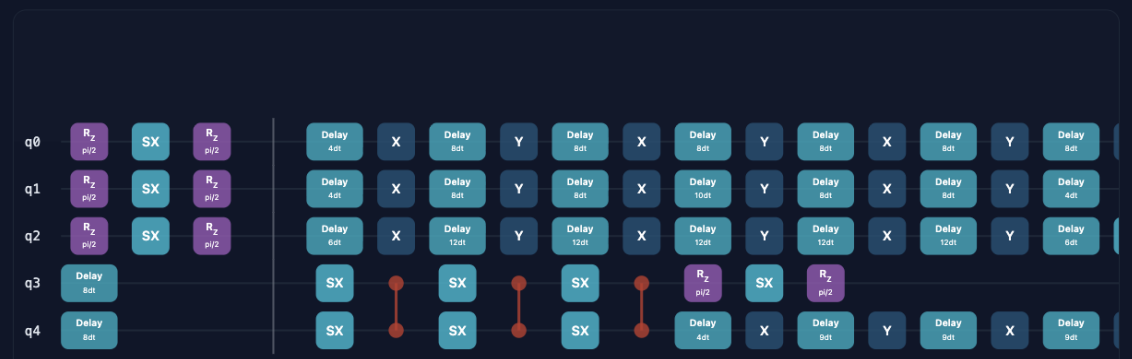
```
OPENQASM 3.0;
include "stdgates.inc";
bit[3] c;
qubit[5] q;
rz(pi/2) q[0];
sx q[0];
rz(pi/2) q[0];
rz(pi/2) q[1];
sx q[1];
rz(pi/2) q[1];
```

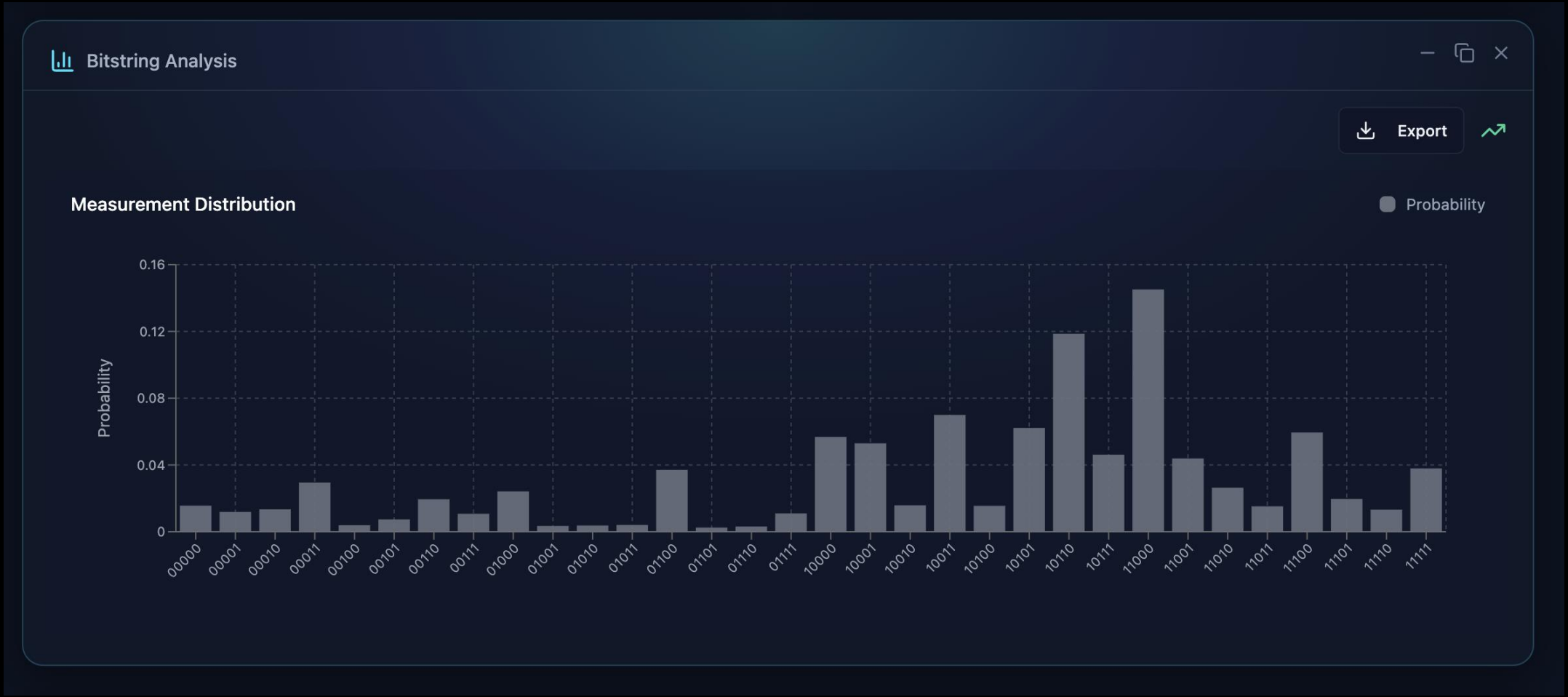
Supports OpenQASM 2.0 and 3.0 formats

Visualize Circuit | Apply DD | Reset QASM

Circuit Visualization 5 qubits 1011 gates

100% Hide idle qubits Run



Tutorial Sample: Improve Shor’s algorithm with dynamical decoupling

Execute the circuits and collect the results